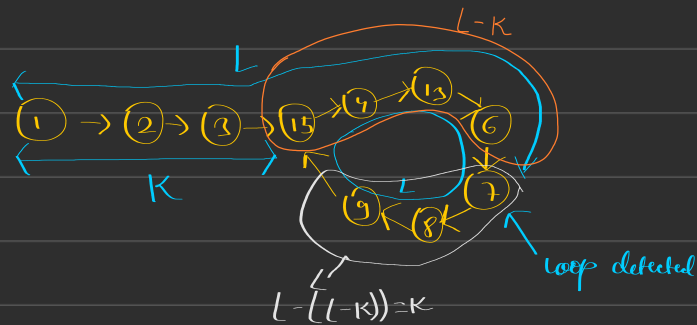
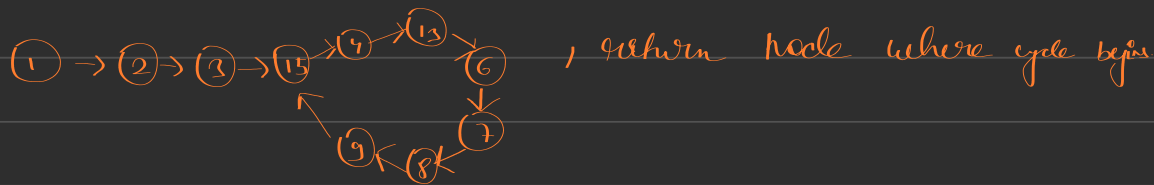


Linked list cycle II:-



At first find the loop's length, it can be found by first getting the item at which both slow and fast matches, then stop fast at that node and move slow by one till $\text{fast} == \text{slow}$, maintain a count.

$$T_c \rightarrow O(n)$$

$$S_c \rightarrow O(1)$$

Length of cycle = L

Outside part of cycle = K

After moving S pointer by L , part of cycle is covered = $L-K$

So, if total is L , and I am already $L-K$ inside, movement needed to complete

$$\text{the cycle} = L - (L-K) = K$$

Now, as the outer part and remaining part is same now both by L , and the point at which they will meet is the starting point of cycle.

$$T_c \rightarrow O(n)$$

$$S_c \rightarrow O(1)$$